

AI Model Gateway Customer Guide

A simple reference for customers who want a multi-model AI platform direction, but may not have a large technical team yet.

Audience	Business and product teams with basic technical awareness
Prototype	Local AI Model Gateway with Qwen / DashScope first
Main idea	One customer API, many model providers behind the scenes
Demo mode	Mock mode first, live Qwen mode only when needed

Executive Summary

An AI Model Gateway lets customers call one simple API while the platform manages model routing, provider differences, usage control, and future fallback logic. The first prototype uses Alibaba Cloud Model Studio / Qwen because it is available now and keeps testing cost low.

- Start with Qwen and mock mode before adding more paid providers.
- Use an OpenAI-compatible API so customers can understand the integration quickly.
- Show routing visually: smart-fast is a customer-facing name that maps to qwen-plus.
- Keep enterprise API Gateway features optional until customer demand is clear.

Layered Architecture

Layer	What It Means	Example
1. Customer Systems	The customer's app, backend, or agent	Website, chatbot, internal tool
2. One Simple API	One standard API format for customers	/v1/chat/completions
3. Gateway Entry Control	Controls who can call and how much they can use	API keys, limits, logs
4. Model Control	Chooses the public model and route	smart-fast -> qwen-plus
5. Provider Adapters	Translates request and response formats	DashScope adapter
6. Model Providers	The real upstream AI model provider	Alibaba Cloud Qwen

The dashboard also shows this as a visual architecture map, so non-technical customers can follow the layers from customer systems to model providers.

How Route Simulation Works

The dashboard explains routing without spending money. In mock mode, the gateway reads the requested model, checks the registry, maps the public model to a real upstream model, and returns a simulated response with a route trace.

```
Customer request: model = smart-fast
Registry lookup: smart-fast maps to qwen-plus
Provider adapter: prepare DashScope-compatible request
```

Mock response: show the route trace without calling Alibaba Cloud

What The Current Prototype Includes

- Visual dashboard at the gateway root URL.
- OpenAPI contract at /openapi.json for customer technical handoff.
- Postman collection at /postman_collection.json for click-through demos.
- Demo bundle manifest at /v1/gateway/demo-bundle for customer presentation flow.
- Handoff checklist endpoint for customer-shareable links, admin-only material, meeting checks, and follow-up actions.
- Customer integration guide at /v1/gateway/integration-guide with safe code examples.
- OpenAI-compatible /v1/chat/completions endpoint.
- Customer API key authentication with Bearer token.
- Basic in-memory request limit.
- Model registry in model_registry.json.
- SQLite request and usage records for restart-safe demo data.
- Provider, customer, model, and request summary endpoints.
- Operational alerts endpoint with simple next steps.
- Incident playbook endpoint with support scenarios and customer-safe wording.
- Support policy endpoint for prototype, pilot, and production support stages.
- Pilot checklist endpoint for before, during, and after a customer trial.
- Pilot scorecard endpoint for deciding discovery, extended pilot, or production hardening.
- Discovery checklist endpoint for customer goals, provider scope, governance, reporting, and production expectations.
- Proposal summary endpoint for customer-facing pilot scope, exclusions, risks, and next steps.
- Deployment readiness endpoint for environment, preflight checks, operations, rollback, and deployment options.
- Migration plan endpoint for phased customer cutover, checklist, rollback, and evidence.
- Production backlog endpoint for prioritized P0, P1, and P2 hardening tasks.
- Executive brief endpoint for non-technical customer stakeholders.
- Roadmap endpoint for prototype, pilot, production hardening, and multi-provider expansion.
- Decision guide endpoint for API Gateway, managed AI Gateway, custom Model Gateway, and OpenRouter-like options.
- FAQ endpoint for common customer questions and objections.
- Demo script endpoint for a 15 minute customer walkthrough.
- Access matrix endpoint for customer and model permissions.
- Provider health endpoint for mock-ready, live-ready, degraded, and not-ready states.
- Provider create, update, and disable actions for provider lifecycle demos.
- Provider contract matrix for adapter differences across Qwen, OpenAI-compatible APIs, Claude, and planned providers.

- Model catalog endpoint for provider status, fallback chain, usage, and pricing metadata.
- Model route create, update, and disable actions for route lifecycle demos.
- Route preview endpoint to dry-run model access, budget, provider readiness, and fallback order.
- Cost estimate endpoint to dry-run token estimate, estimated cost, and budget impact.
- Customer reports endpoint for request count, errors, token usage, and budget state.
- Commercial policy endpoint for pricing assumptions, budget behavior, invoice preview limits, and exclusions.
- Procurement pack endpoint for vendor review tracks, evidence documents, approval owners, and red lines.
- Business case endpoint for value hypotheses, pilot metrics, ROI inputs, and decision options.
- Implementation plan endpoint for delivery phases, duration ranges, roles, risks, and acceptance evidence.
- Alternatives pack endpoint for direct provider, API Gateway, managed AI Gateway, OpenRouter-like, and custom gateway comparison.
- SOW draft endpoint for proposed scope, deliverables, exclusions, milestones, acceptance criteria, and open contract items.
- Workshop agenda endpoint for attendees, pre-work, meeting flow, questions, demo order, outputs, and follow-up actions.
- Decision log endpoint for workshop decisions, owners, evidence, open questions, next actions, and meeting notes.
- Follow-up email endpoint for customer recap, links, next actions, boundaries, and internal checklist.
- Pilot kickoff endpoint for pilot goals, attendees, checklist, agenda, technical start, success metrics, and risks.
- Pilot review endpoint for stop, extend, or production hardening decision support.
- Production transition endpoint for moving from pilot review to hardening work without over-promising production readiness.
- Operating review endpoint for recurring customer health, usage, incidents, budget, change, and expansion decisions.
- Provider expansion endpoint for adding OpenAI, Claude, Xiaomi, OpenRouter-like, or other providers with gates and tests.
- Customer expansion endpoint for growing one customer from pilot to more teams, workflows, models, providers, or traffic.
- Change request preview endpoint for dry-run impact, approval owner, evidence, and rollback before saving a change.
- Request activity endpoint with simple filters for troubleshooting.
- Request detail lookup using `gateway.request_id` returned in chat responses.
- Config check endpoint for demo keys and missing production settings.
- Production readiness endpoint for plain-English go-live gaps.
- Data governance endpoint for prompt handling, logs, retention, customer keys, and provider secrets.
- Security review endpoint for threat model, current controls, production controls, and go-live security gates.
- Operations runbook endpoint for SLO-style targets, daily checks, alert actions, and owners.
- Evaluation plan endpoint for model quality, reliability, latency, cost, safety, fallback, and scorecards.
- Structured route decision summary for support explanations.
- Local safety preview for obvious sensitive data.

- Customer key issue preview for safe onboarding demos.
- Customer key create, rotate, and disable actions for lifecycle demos.
- Audit events for customer key lifecycle actions.
- Customer default policy for automatic routing presets.
- Invoice preview with JSON and CSV output.
- Capability routing control for streaming and tools.
- Request-level fallback controls for routing demos.
- Request-level provider allow-list for routing control demos.
- Request-level route strategy for registry, lowest cost, fastest, and healthiest routing.
- Named policy presets for common routing controls.
- Mock OpenAI-style tool call response for agent demos.
- Simple customer plans with token and cost budgets.
- Bring Your Own Key mapping through environment variables.
- Provider adapter scaffolds for OpenAI-compatible APIs and Claude-style APIs.
- Automated mock regression test that does not spend provider credits.
- Mock mode for demos and planning.
- Live Qwen mode when DASHSCOPE_API_KEY is available.

Main Technical Difficulties

Difficulty	Why It Matters	Prototype Status
Provider format differences	Each provider may use different request and response details.	Handled first for DashScope-compatible chat
API handoff	Customer technical teams need a clear API contract.	OpenAPI contract endpoint
Demo handoff	Customer technical teams may want to click through requests.	Postman collection endpoint
Demo flow	Business and technical users need to know what to look at.	Demo bundle manifest endpoint
Customer handoff	Teams need to know what can be shared and what stays add-on only.	Handoff checklist endpoint
Customer integration	A customer technical team needs safe examples for their own key integrations.	Customer integration guide endpoint
Streaming	Chat UIs and agents often need incremental tokens.	Mock streaming included; live provider differences
Tool calling	OpenAI, Claude, and Qwen may differ in tool format.	Mock tool_calls plus basic normalization
Token usage	Billing and quota require reliable usage data.	Stored in SQLite
Operational alerts	Non-technical users need a clear action list.	Alerts with severity, area, and next step
Incident response	Support teams need a shared script when requests fail.	Playbook with signals, steps, and customer wording
Support policy	Customers need to know what support is promised at each stage.	Prototype, pilot, and production support guide
Pilot planning	A customer trial needs scope, roles, success criteria, and exit criteria.	Pilot checklist endpoint
Pilot scoring	Teams need to know whether the pilot worked.	Scorecard with score, decision, weak criteria, and notes
Customer discovery	Broad gateway ideas need clear scope before implementation.	Discovery checklist with questions, evidence, and notes
Proposal summary	Customers need a simple scope note after discovery.	Customer-facing pilot scope, exclusions, risks, and notes
Executive communication	Non-technical stakeholders need a short business summary.	Executive brief endpoint
Roadmap planning	Customers need to see the path from demo to production.	Roadmap endpoint
Build or buy decision	Customers need to compare gateway options clearly.	Decision guide endpoint
Customer objections	Sales and support need consistent answers to common questions.	FAQs endpoint
Demo delivery	A presenter needs a simple meeting flow, talk track, and likely questions.	Demo script endpoint
Production readiness	Customers need to understand why demo-ready is not production-ready.	Production readiness report
Deployment readiness	Customer technical teams need to know what must be confirmed before opening it.	Required before opening health checks, rollback, and notes
Customer migration	Customers need a low-risk path from direct provider calls to the gateway.	Phase gateway plan and rollback checklist
Production backlog	Teams need to convert gaps into fundable engineering tasks.	R3, P1, and P2 hardening backlog
Data governance	Customers need to know what happens to prompts, logs, and data.	Privacy and data governance review
Security review	Customers will ask what can go wrong and what controls exist.	Threat model, current controls, and go-live gates

Operations runbook	Customers need to know who watches the gateway after a pilot starts.	Pilot status checks, owners, and alert actions
Model evaluation	Customers will ask how the gateway chooses a better model.	Evaluation plan, sample prompts, and scorecards
Access control	Each customer may be allowed to use different models.	Access matrix by customer and model
Provider health	Customers need to know whether a provider can serve traffic.	Readiness view based on config and recent traffic
Provider lifecycle	Teams need to add and stop providers safely.	Local JSON-backed create, update, and disable a
Provider contracts	Teams need to see why each provider needs adapter tests.	Contract matrix for provider differences
Model catalog	Customers need to understand public model names and routes.	Catalog with provider status, fallback chain, usage
Model route lifecycle	Teams need to change public model routes safely.	Local JSON-backed create, update, and disable a
Route preview	Sales and support need to explain a route before spending money.	Dry-run endpoint and dashboard preview
Route decision summary	Customers may ask why a route was chosen.	Plain-language summary and reasons
Safety preview	Customers may worry about secrets in prompts.	Local preview for obvious emails, phone numbers
Provider routing control	Some customers may want only approved providers for a region.	gateway_allowed_providers filters route candidate
Route strategy	Customers may want cost, latency, or health-aware routing.	Request-level route strategy reorders candidates
Policy presets	Non-technical users need simple policy names.	gateway_policy applies named routing presets
Customer default policy	Customers may not want to send routing controls every time.	Customer config can set a default policy
Capability routing	Requests may need streaming, tools, or other abilities.	gateway_required_capabilities filters route candid
Cost estimate	Customers need budget planning before live calls.	Dry-run estimate for tokens, cost, and budget imp
Commercial policy	Customers need to know what is an estimate and what requires a contract.	Pricing assumptions, budget rules, and exclusions
Procurement pack	Customers may need procurement, legal, IT, security, and finance reviews.	Review tracks evidence documents, approval ow
Business case	Business sponsors need to explain why a pilot is worth doing.	Value hypotheses, pilot metrics, ROI inputs, and c
Implementation plan	Customers need to know how the idea becomes real work.	Delivery phases, rough duration ranges, roles, ris
Alternatives pack	Customers may ask why not use OpenRouter, Vercel AI Gateway, Alibaba Cloud AI Gateway, or other	Buildup, Alibaba Cloud AI Gateway, custom gatewa
SOW draft	Customers may need a safe project scope draft before legal review.	Scope, deliverables, exclusions, milestones, acce
Workshop agenda	The first customer meeting needs structure and clear output.	Attendees, pre-work, agenda, questions, demo or
Decision log	Workshop decisions need owners and follow-up evidence.	Decision records, open questions, next actions, st
Follow-up email	After a workshop, customers need a simple recap and safe subject lines.	Subject options, email draft, links, actions, bound
Pilot kickoff	A demo is not enough; a pilot needs owners, tests, metrics, and operating procedures.	Goals, attending, checklist, agenda, technical sta
Pilot review	A pilot should end with a clear decision, not another vague question.	Evidence, go/no-go checks, stop/extend/harden o
Production transition	Pilot success does not mean production approval.	Transition trigger, phases, role handoff, P0 contro

Operating review	A gateway needs ongoing ownership after launch or limited production signals, customer snapshots, decisions	Customer signals, customer snapshots, decisions
Provider expansion	Adding providers is governance work, not just another API key	Provider candidates, approval gates, tests, phase
Customer expansion	One successful pilot should not become unlimited usage as a stage	Stages, gates, decisions, snapshots, questions, a
Change request preview	Config changes can affect customers, budget, support, and fallback	Config impact, approval owner, checklist, eviden
Customer onboarding	New customers need keys, limits, and model access.	Key issue preview creates a safe config snippet
Key lifecycle	Customers need key rotation and disable workflows.	Local JSON-backed create, rotate, and disable ac
Audit trail	Teams need to know who changed customer access.	Audit events for customer lifecycle actions
Customer self-service	Customers need to see their own access, usage, and budget	Customer self view with no provider secret expos
Invoice preview	Customers need a simple billing story.	Estimated invoice preview with CSV export
Customer reporting	Customers need a simple usage and budget story.	Per-customer report endpoint and dashboard car
Request troubleshooting	Support teams need to see what happened to a request.	Filtered request activity feed
Request detail	Support teams need one-request lookup.	gateway.request_id and request detail endpoint
Fallback	Retrying another model needs clear business rules.	Registry fallback plus request-level controls
Customer key management	Each customer needs limits, logs, and access control.	Minimum version plus BYOK mapping
Cost control	Different models have different prices and limits.	Simple token and cost budgets included

Why The Test Script Matters

The repository includes `test_gateway.py`. It starts the gateway in mock mode and checks API keys, model listing, route tracing, fallback, model access rules, token budget blocking, SQLite logs, and status output without leaking provider keys.

API Contract

`/openapi.json` exposes an OpenAPI contract for the prototype. It describes the customer API, admin control-plane API, bearer key security schemes, and the main endpoint groups. `/postman_collection.json` exposes a ready-to-import Postman collection with `base_url`, `gateway_api_key`, and `admin_api_key` variables.

`/v1/gateway/demo-bundle` lists the dashboard, customer self view, OpenAPI contract, Postman collection, PDF guide, recommended demo order, safe curl examples, and production notes. `/v1/gateway/handoff-checklist` explains what can be shared with customers, what should stay admin-only, meeting checks, follow-up actions, and the rule that provider API keys stay private. The dashboard also has a Customer handoff package section that links the main artifacts in one place and an Integration command starter with local curl commands. `/v1/gateway/integration-guide` lets a customer use their own key to see allowed models, a recommended first model, curl, Python, JavaScript, streaming examples, and a go-live checklist without exposing provider secrets. `/v1/gateway/sdk-starter` adds a `.env` template, starter Python and JavaScript files, first-run commands, common error fixes, and a handoff checklist. These help a business team understand the story first and help a customer technical team import or click through the prototype quickly.

Handoff Checklist

`/v1/gateway/handoff-checklist` is an admin-only customer meeting checklist. It separates customer-shareable material from internal readiness material. It covers the business overview, customer technical handoff, pilot decision package, internal readiness package, meeting checks, and follow-up actions.

Group	What It Means
Business overview	Links for non-technical customer explanation.
Customer technical handoff	API contract, Postman, integration guide, and SDK starter.
Pilot decision package	Pilot checklist, scorecard, customer reports, and success summary.
Internal readiness package	Production readiness, deployment readiness, backlog, change plan, and data governance.

Admin Summary Endpoints

The prototype exposes local admin JSON views for provider status, operational alerts, access matrix, provider health, model catalog, route preview, customer reports, request activity, request detail lookup, usage by customer, usage by model, and request summaries. These make the control layer easier to explain. The local demo uses a separate admin key. In production, this key should be changed and protected.

Operational Alerts

`/v1/gateway/alerts` combines config warnings, provider readiness, customer budget state, and recent request errors. Each alert includes severity, area, message, and a simple next step. This helps non-technical users understand what needs attention before a live demo.

Incident Playbook

`/v1/gateway/incident-playbook` explains common failure scenarios in simple English. It covers provider readiness, recent request errors, customer budget blocks, demo key or secret risks, and provider contract gaps. Each scenario includes triggers, signals to check, current evidence, operator steps, and customer-safe wording. It is not a legal SLA, but it helps sales, support, and technical teams explain issues consistently.

Support Policy

`/v1/gateway/support-policy` explains prototype, technical pilot, and production target support expectations. It defines P1, P2, and P3 severity, first checks, escalation path, and customer-safe wording. It clearly says the current prototype is not a legal production SLA. This helps customers understand what can be promised now and what needs a real contract later.

Pilot Checklist

`/v1/gateway/pilot-checklist` helps prepare a small customer trial. It covers recommended scope, roles, before-pilot checks, during-pilot checks, after-pilot review, success criteria, and the stop, extend, or productionize decision. It helps keep the pilot small, honest, and measurable.

Onboarding Plan

`/v1/gateway/onboarding-plan` turns the demo into a simple customer pilot path. It explains Day 0 customer alignment, Day 1 safe access setup, Day 2 mock technical test, Day 3 route and provider review, Day 4 usage, cost, and support review, and Day 5 pilot decision. Each step has an owner, actions, evidence, and an exit check, so non-technical

customers can understand what happens after the demo.

Executive Brief

`/v1/gateway/executive-brief` gives a short business summary for non-technical stakeholders. It explains the one-sentence value, why the gateway matters, what the demo proves, what is not production-ready yet, the recommended customer story, pilot recommendation, support and readiness position, and next step.

Roadmap Endpoint

`/v1/gateway/roadmap` explains the path from prototype to production. It includes prototype explanation, technical pilot, production hardening, and multi-provider expansion. Each phase has a goal, deliverables, exit criteria, and main risks. This helps customers understand next steps without jumping too quickly into production promises.

Decision Guide

`/v1/gateway/decision-guide` compares normal API Gateway, managed AI Gateway, custom Model Gateway, and OpenRouter-like marketplace options. The dashboard also renders this as Gateway options comparison cards. It explains what each option is good at, what it does not solve, and when to choose it. This directly answers the early question: is an API Gateway enough?

Customer FAQ

`/v1/gateway/faq` answers common customer questions about API Gateway, Qwen, OpenRouter, prompt and key exposure, cost control, provider failure, adding more providers, pilot next steps, and production readiness. It helps sales, support, and technical teams answer in the same simple language.

Demo Script

`/v1/gateway/demo-script` gives a 15 minute customer walkthrough. The dashboard also shows this as Presenter mode. It tells the presenter what to say first, which endpoint to show, how to explain routing, how to answer common objections, and how to close with a small pilot. This helps non-technical customers understand the project without reading code.

Customer Self View

`/v1/gateway/me` lets a customer use their own gateway key to see their own plan, allowed models, budget state, usage, recent requests, and invoice preview. It does not expose raw customer API keys or provider API keys. This is useful when the customer asks: what can I use, and how much have I used?

Access Matrix

`/v1/gateway/access-matrix` shows which customers can use which public models, why access is allowed or blocked, the customer's budget state, and the provider readiness for each model. This helps explain customer-level API control.

Provider Health

`/v1/gateway/provider-health` answers a simple question: can this provider serve traffic now? It checks enabled models, live key readiness, customer BYOK readiness, recent errors, and average latency. In mock mode, `ready_mock` means the provider can be explained without spending credits. It does not mean the live provider key is ready.

Provider Lifecycle

The prototype can create, update, and disable providers. The dashboard has a Provider lifecycle panel for these demos. A provider defines the upstream API type, base URL, and API key environment variable name. Disabling a provider also disables active model routes that point to it, so the local registry stays valid. These admin actions update `model_registry.json`, reload the local runtime, and write audit events. The prototype stores the provider key environment variable name, not the provider secret value. Production should use a database, secret manager, approval workflow, readiness checks, and rollout controls.

Provider Contract Matrix

`/v1/gateway/provider-contracts` explains why an AI Model Gateway is more than a normal API Gateway. It compares OpenAI-compatible providers, Alibaba Cloud Model Studio compatible mode, Anthropic Claude, and planned Xiaomi or other local model providers. It shows auth, endpoint path, request shape, response shape, streaming, tool calling, usage fields, adapter status, and remaining contract gaps.

Model Catalog

`/v1/gateway/model-catalog` explains what each public model name means. It shows the upstream model, provider status, fallback chain, capabilities, pricing metadata, request count, errors, tokens, and estimated cost. This helps customers understand that smart-fast can be a simple public name while the gateway manages the real provider route behind it.

Model Route Lifecycle

The prototype can create, update, and disable public model routes. The dashboard has a Model route lifecycle panel for these demos. A route defines the customer-facing model name, provider, upstream model, fallback models, capabilities, and pricing metadata. These admin actions update `model_registry.json`, reload the local runtime, and write audit events. Production should add approval workflow, version history, rollback, and staged rollout.

Route Preview

`/v1/gateway/route-preview` is a dry run. It does not call the provider. It shows whether a customer can use a model, whether budget is available, which model is primary, which models are fallback routes, which provider would handle the request, and whether that provider is ready.

Route Decision Summary

Route preview and chat responses include `route_decision`. It explains the requested public model, selected upstream model, provider, fallback policy, provider controls, capability controls, and simple reasons for the routing decision. This is useful when a customer asks why a request used a certain model or provider.

Safety Preview

`/v1/gateway/safety-preview` checks for obvious sensitive data before a provider call. It can detect examples such as email addresses, possible phone numbers, possible API keys, and secret assignments. Chat requests can also use `gateway_safety_check` to return safety metadata, `gateway_block_sensitive` to block detected sensitive data, or `gateway_redact_sensitive` to replace detected sensitive data before the provider call. This is useful for explaining gateway guardrails, but it is not a full DLP or compliance system.

Provider Routing Control

The prototype supports `gateway_allowed_providers`. This lets a request say: only use these providers for this route. For example, route preview can allow only dashscope. If no route matches the allowed provider list, the gateway returns a clear error. This helps customers understand that the gateway is a control layer, not only a normal API proxy.

Routing Strategy

The prototype supports `gateway_route_strategy`. Supported values are `registry`, `lowest_cost`, `fastest`, and `healthiest`. The strategy reorders route candidates before the provider call. Route preview and chat responses include `candidate_scores` in `route_decision`. This helps explain cost-aware, latency-aware, and health-aware routing. Production should use stronger cost data, latency windows, provider SLAs, and customer policy rules.

Policy Presets

The prototype supports `gateway_policy`. A policy preset is a short name for common routing controls. Current presets include `balanced`, `lowest_cost`, `fastest`, and `tool_ready`. `/v1/gateway/policy-presets` lists the presets. Explicit request controls override preset controls. This helps non-technical customers choose a simple policy name instead of many technical fields.

Customer Default Policy

A customer can have `default_policy` in `customer_keys.json`. If a request does not send `gateway_policy`, the gateway uses the customer default. If a request sends `gateway_policy`, the request value wins. This lets a customer use one gateway key and one default routing behavior without sending routing controls every time.

Capability Routing Control

The prototype supports `gateway_required_capabilities`. This lets route preview or a real request require abilities such as streaming or tools. `stream=true` requires streaming. A `tools` request requires tools. If no route supports the required ability, the gateway returns a clear `no_capability_route` error. This explains why model metadata matters in a real gateway.

Cost Estimate

`/v1/gateway/cost-estimate` is a dry run. It does not call the provider. It estimates prompt tokens, completion tokens, estimated cost, and remaining budget after the estimate. This is useful for planning, but real provider token usage can differ.

Customer Key Issue Preview

`/v1/gateway/key-issue-preview` creates a safe customer onboarding package. It returns a generated gateway API key, a masked key for display, a customer config snippet, allowed models, limits, and next steps. It does not save the customer. In production, this would connect to a real customer database and secret manager.

Customer Key Lifecycle

The prototype can also create a customer, rotate a customer gateway key, and disable a customer. The dashboard has a Customer lifecycle panel for these demos. These admin actions update `customer_keys.json` and reload the local runtime. After rotation, the old key stops working. After disable, the customer key stops working. This is useful for demos, but production should use a database, audit logs, approval workflow, and a secret manager.

Audit Events

/v1/gateway/audit-events records admin lifecycle actions such as customer created, key rotated, customer disabled, provider changed, and model route changed. The dashboard also has an Audit timeline section for recent changes. Each event includes actor, action, target customer, provider, or model route, time, and safe details such as masked keys and key hashes. It does not store full gateway API keys. This helps explain control history, but it is not a full compliance audit system.

Commercial Policy

/v1/gateway/commercial-policy explains how to talk about pricing, budgets, invoice previews, and commercial boundaries before a real contract exists. It covers what can be shown now, what needs contract approval later, request limit behavior, token budget behavior, cost budget behavior, invoice preview limits, approval questions, and prototype exclusions. It is not a legal quote, tax invoice, payment system, or audited billing ledger.

Area	Plain-English meaning
What can be shown now	Estimated usage, estimated cost, budgets, invoice preview, and CSV export.
Contract later	Final pricing, payment terms, overage behavior, refunds, tax, and legal invoice fields.
Budget behavior	Request, token, and cost limits can block or warn in the prototype.
Exclusions	No legal quote, payment collection, refund workflow, or audited billing ledger yet.

Procurement Pack

/v1/gateway/procurement-pack helps a customer share the idea with procurement, legal, IT, security, and finance before a pilot or purchase discussion. It explains what the prototype is, what it is not, review tracks, evidence documents, approval owners, red lines, and meeting questions. It does not replace a signed contract, legal security attestation, production SLA, final price quote, or tax invoice.

Area	Plain-English meaning
Review tracks	Business, IT, security, data, finance, legal, and customer technical teams know what to check.
Evidence documents	The pack points to OpenAPI, Postman, security review, data governance, commercial policy, and
Approval owners	Each owner knows what they must approve before production.
Red lines	The team knows which promises should wait for production approval.

Business Case

/v1/gateway/business-case helps a business sponsor explain why the gateway may be worth a pilot. It shows a simple value story, value hypotheses, pilot metrics from local usage records, ROI inputs the customer must provide, decision options, and what the prototype does not claim. It is not a formal ROI model and does not promise guaranteed savings, final production price, production SLA, security certification, or legal billing output.

Area	Plain-English meaning
------	-----------------------

Value hypotheses	Integration work, provider control, budget visibility, lower change risk, and clearer approval can be
Pilot metrics	Local requests, tokens, estimated cost, active customers, and readiness status are visible.
ROI inputs	The customer must provide engineering cost, current integration effort, risk cost, and finance assu
Decision options	Stop after demo, run a small pilot, or harden for production.
Not claimed	No guaranteed savings, formal ROI, final price, SLA, certification, or legal invoice.

Implementation Plan

/v1/gateway/implementation-plan helps a customer understand what it would take to move from idea to pilot to production. It explains delivery phases, rough duration ranges, main work, roles, estimate assumptions, delivery risks, acceptance evidence, and what should not be promised too early. It is not a fixed delivery quote and does not promise final price, guaranteed delivery date, production SLA, security certification, legal approval, or provider cost guarantee.

Area	Plain-English meaning
Delivery phases	Discovery, prototype demo, controlled pilot, production hardening, and launch.
Roles	Business, gateway, platform, security, data, finance, legal, and customer technical owners are nar
Assumptions	The first pilot starts narrow: one customer team and one primary model route.
Risks	Provider differences, data policy, budget approval, support ownership, and scope growth are listed
Acceptance evidence	Each stage points to endpoints that prove readiness or remaining gaps.

Alternatives Pack

/v1/gateway/alternatives-pack compares direct provider integration, normal API Gateway, Alibaba Cloud AI Gateway, Vercel AI Gateway, OpenRouter-like platforms, and a custom AISmallRouter-style Model Gateway. It explains when each option fits, what tradeoffs exist, and why a custom gateway should only be built when customer-owned controls, custom routing, private reports, or special rollout rules matter. It is not a live vendor benchmark, legal recommendation, final procurement decision, or promise to build a full OpenRouter clone.

Option	Plain-English fit
Direct provider	Best when there is one provider and low need for central control.
Normal API Gateway	Best for HTTP traffic control, WAF, rate limits, and API logs.
Managed AI Gateway	Best when hosted platform ownership and managed provider routing are acceptable.
OpenRouter-like	Best when broad public model access matters more than private gateway ownership.
Custom Model Gateway	Best when customer-owned controls, reports, audit, and rollout rules matter.

SOW Draft

/v1/gateway/sow-draft helps turn the gateway discussion into a clear project scope. It includes proposed scope, deliverables, out-of-scope items, milestones, acceptance criteria, assumptions, customer inputs, risk controls, and open items before signature. It is only a starting point and is not a signed contract, final quote, tax invoice, production SLA, security certification, or legal approval.

Area	Plain-English meaning
Scope	What the first pilot SOW can safely include.
Deliverables	Prototype, technical handoff, business materials, control-plane explanation, and production planning.
Out of scope	Marketplace, full SLA, legal certification, final billing engine, many providers, and guaranteed provi
Milestones	Scope confirmation, prototype walkthrough, controlled pilot setup, and hardening decision.
Acceptance	Evidence-based checks such as API calls, route explanation, reports, and no secret exposure.

Workshop Agenda

/v1/gateway/workshop-agenda helps run the first customer meeting. It lists who should attend, what to prepare, agenda topics, questions to ask, demo order, expected outputs, follow-up actions, and red flags. It helps turn a broad AI gateway idea into one use case, one provider path, one pilot owner, and one next decision. It is a meeting guide, not a contract, quote, SLA, or production approval.

Area	Plain-English meaning
Attendees	Business, technical, security, procurement, finance, gateway, and support owners know why they
Agenda	Business goal, current pain, gateway demo, options, controls, pilot shape, and next steps.
Questions	Use case, first provider, logging, budgets, approval teams, and success criteria.
Outputs	Confirmed use case, owner, provider path, data answer, pilot criteria, and next artifact.
Follow-up	Each owner has a concrete action and evidence endpoint.

Decision Log

/v1/gateway/decision-log records what was decided after a customer workshop, who owns each decision, which evidence endpoints support the decision, what remains open, and what next actions should happen. It also includes a simple meeting note template. It is a planning record, not a contract, legal approval, invoice, SLA, or security certification.

Area	Plain-English meaning
Decision records	First use case, first provider, mock/live, build/buy, success criteria, and proposal/SOW next step.
Open questions	Sponsor, use case, live testing, logging, budget, approvals, and decision date.
Next actions	Gateway, sponsor, technical, security, and finance/procurement owners each get an action.

Notes template	Simple structure for sending clean meeting notes.
Boundary	Not customer approval, signed contract, legal acceptance, production approval, or final billing decision.

Follow-up Email

`/v1/gateway/follow-up-email` helps write the customer email after a workshop. It includes subject options, a customer email draft, recap points, links to include, decision summary, next actions, internal checklist, and things not to claim. It is a communication draft, not a signed contract, final quote, invoice, SLA, or security approval.

Area	Plain-English meaning
Email draft	A simple thank-you message with the gateway idea, pilot path, and open items.
Links	Dashboard, OpenAPI, Postman, proposal summary, SOW draft, and decision log.
Next actions	Owner-based actions copied from the decision log.
Do not claim	No provider keys, unapproved production dates, final price, or security certification wording.
Boundary	Not a contract, final quote, invoice, SLA, security approval, or OpenRouter clone.

Pilot Kickoff

`/v1/gateway/pilot-kickoff` helps start a controlled customer pilot after the customer agrees to try the gateway. It includes kickoff goals, required attendees, pre-kickoff checklist, meeting agenda, technical start points, success metrics, operating rhythm, risks to watch, and handoff actions after kickoff. It is a pilot planning pack, not production approval, a signed SOW, a final SLA, or a security certification.

Area	Plain-English meaning
Goals	Confirm use case, owner, mock/live mode, model access, budget, data rules, and decision date.
Attendees	Sponsor, technical contact, security/data owner, support owner, and gateway owner.
Technical start	Customer gateway key, integration guide, SDK starter, OpenAPI, and Postman.
Success metrics	First call, route understanding, usage visibility, support traceability, and decision readiness.
Boundary	Not production approval, signed SOW, final SLA, security certification, or unlimited provider coverage.

Pilot Review

`/v1/gateway/pilot-review` helps review a pilot and decide whether to stop, extend the pilot, or move toward production hardening. It includes review inputs, meeting agenda, decision options, evidence summary, customer report snapshot, go/no-go checks, recommended decision wording, follow-up outputs, and things not to claim. It is decision support, not production approval, a signed SOW, legal approval, a final quote, or a security certification.

Area	Plain-English meaning
------	-----------------------

Review inputs	Pilot kickoff, scorecard, customer reports, request activity, readiness, and backlog.
Decision options	Stop, extend pilot, or move to production hardening.
Go/no-go checks	Sponsor, use case, API test, usage visibility, data rules, security P0, backlog, and commercial pat
Recommended wording	Pilot can prove the direction; production needs hardening and approval.
Boundary	Not production approval, customer acceptance, final quote, signed contract, or security certification

Production Transition

/v1/gateway/production-transition helps move from a successful pilot review into production hardening. It includes the transition trigger, hardening phases, role handoff, P0 controls before live traffic, P1 items for limited production, customer questions before SOW, internal kickoff agenda, customer-safe message, and things not to claim. It is a transition plan, not production approval, a signed SOW, final price, legal approval, or security certification.

Area	Plain-English meaning
Trigger	Use when the pilot has value but production controls are not approved yet.
Phases	Confirm scope, close P0 controls, prepare deployment, approve gates, and start limited traffic.
Role handoff	Sponsor, technical, gateway, platform, security/data, support, and finance owners.
P0 controls	Secrets, storage, data governance, monitoring, and change control before live traffic.
Boundary	Not production approval, signed SOW, final price, legal approval, security certification, or SLA.

Operating Review

/v1/gateway/operating-review helps run weekly or monthly reviews after the gateway is being tested or used. It includes review cadence, meeting agenda, operating signals, customer snapshots, expand/hold/fix/pause decision options, owner follow-ups, review outputs, and things not to claim. It is an operating review guide, not an SLA report, legal audit, final invoice, or security certification.

Area	Plain-English meaning
Cadence	Daily during first week, weekly during pilot or limited production, monthly after stable usage.
Signals	Customer health, alerts, usage totals, request count, support stage, and operating stage.
Decisions	Expand carefully, hold steady, fix blockers first, or pause/roll back.
Owners	Gateway, support, customer success, security/data, finance, and platform owners.
Boundary	Not an SLA report, legal audit, security certification, final invoice, or uptime guarantee.

Provider Expansion

/v1/gateway/provider-expansion helps plan how to add more model providers without promising an uncontrolled marketplace. It includes expansion principles, provider candidates, approval gates, test matrix, expansion phases, customer questions, current provider context, and things not to claim. It is an expansion planning pack, not a live vendor benchmark, procurement approval, provider certification, final price, or full OpenRouter clone promise.

Area	Plain-English meaning
Principles	Add providers one at a time, with one use case, one model alias, and provider keys kept private.
Candidates	Qwen first, then OpenAI-compatible, Claude, Xiaomi or regional providers, or OpenRouter-like options.
Approval gates	Business value, technical contract, security/data, cost, support, and customer communication.
Tests	Models, route preview, mock chat, live smoke test, streaming, tools, cost, fallback, and rollback.
Boundary	Not a full OpenRouter clone, live benchmark, procurement approval, provider certification, or unlimited usage.

Customer Expansion

/v1/gateway/customer-expansion helps expand one customer from a small pilot to more teams, workflows, models, providers, or traffic. It includes expansion principles, stages, control gates, decisions, customer snapshot, access snapshot, customer questions, and things not to claim. It is an expansion plan, not a signed contract, unlimited usage approval, production SLA, final price, or security certification.

Area	Plain-English meaning
Principles	Expand one dimension at a time and set budget, access, data, and support rules first.
Stages	Single workflow pilot, more users, more workflows, more models/providers, then broader rollout.
Control gates	Customer access, budget, data handling, support, quality, and provider route.
Decisions	Expand usage, add workflow, add provider or model, or hold/reduce scope.
Boundary	Not signed contract, unlimited usage approval, production SLA, final price, or security certification.

Invoice Preview

/v1/gateway/invoice-preview uses local usage records to create an estimated billing preview. It shows requests, errors, prompt tokens, completion tokens, total tokens, estimated cost, remaining budget, usage by model, and usage by provider. It can return JSON or CSV. It is not a legal invoice, but it helps explain how customer billing reports could work later.

Customer Reports

/v1/gateway/customer-reports shows one report per customer. It includes request count, error count, token usage, remaining budget, models used, providers used, and recent requests. This helps explain that a gateway is also a control and reporting layer, not only a model router.

Customer Success Summary

`/v1/gateway/customer-success` turns customer usage data into account health. It shows healthy, watch, and at-risk customer counts, health status per customer, risk reasons, recommended follow-up action, simple business metrics, meeting questions, and evidence endpoints for support. This helps sales, support, and customer success teams decide who needs attention before the next customer call.

Pilot Scorecard

`/v1/gateway/pilot-scorecard` helps decide whether a customer pilot should stay in discovery, continue testing, or move toward production hardening. It scores first request completion, request tracing, error rate, budget state, model access, and production gap acknowledgement. Each customer gets a score, decision, metrics, weak criteria, and next action.

Request Activity

`/v1/gateway/request-activity` shows recent gateway decisions. It can filter by customer, model, provider, error code, status, and limit. This helps explain which customer sent a request, which public model was requested, which upstream model was used, which provider handled it, and whether the request succeeded or failed.

Request Detail

Every successful chat response includes `gateway.request_id`. `/v1/gateway/request-detail` can look up one request by that id. It shows the customer, public model, upstream model, provider, outcome, latency, and nearby usage record when available.

Config Check

The prototype includes `/v1/gateway/config-check`. It warns about demo admin keys, demo customer keys, missing live provider keys, and direct secret values in local JSON files. This is not a full security audit, but it is a useful readiness checklist for customer demos.

Production Readiness Report

`/v1/gateway/production-readiness` explains go-live gaps in plain English. The dashboard also shows this as Production readiness cards. It groups the work into security, provider readiness, customer controls, routing and fallback, observability, billing, and documentation handoff. Each area has a status, evidence, and next step. This helps a customer understand why a prototype can be demo-ready but still not production-ready.

Deployment Readiness Guide

`/v1/gateway/deployment-readiness` explains how to move from local demo to a controlled pilot or production deployment. It covers local demo, live Qwen test, technical pilot, and production target stages. It also lists required environment variables and files, preflight checks, operational health checks, rollback plan, and deployment options. It is not a one-command production deploy.

Area	What it explains
Stages	Local demo, live Qwen test, technical pilot, and production target.
Environment	Admin key, customer keys, DASHSCOPE_API_KEY, registry, customer config, and logs.
Preflight	Admin key, provider key strategy, customer keys, data policy, and readiness blockers.

Operations	Health check, model list, mock chat, route preview, and provider health.
Rollback	Config versioning, audit events, database rollback, and fallback route plan.

Customer Migration Plan

`/v1/gateway/migration-plan` explains how a customer can move from direct model provider calls to one gateway API without switching everything at once. It covers current-state discovery, shadow gateway setup, mock and evaluation testing, limited live pilot, gradual cutover, production decision, cutover checklist, rollback plan, and evidence endpoints. It is a cutover plan for customer conversations, not a one-click migration tool.

Phase	Purpose
Current-state discovery	Understand current provider calls, prompts, models, owners, and data rules.
Shadow gateway setup	Create public model names and customer key without changing production traffic.
Mock and evaluation test	Show routing, fallback, cost estimate, safety, and model evaluation evidence.
Limited live pilot	Send small approved traffic through the gateway while old provider path remains available.
Gradual cutover	Increase traffic only after error, latency, budget, and support checks pass.
Production decision	Choose stop, extend pilot, or fund production hardening.

Production Hardening Backlog

`/v1/gateway/production-backlog` turns prototype gaps into prioritized engineering work. It groups work into P0 production blockers, P1 pilot and limited-production hardening, and P2 scale or marketplace expansion. It covers secrets, storage, data governance, operations, change control, provider contracts, billing, tenant controls, marketplace planning, and deployment automation.

Priority	Meaning
P0	Must be done before production customer traffic.
P1	Needed for a serious pilot or limited production.
P2	Useful for scale, automation, or later multi-provider expansion.

Production Launch Plan

`/v1/gateway/launch-plan` converts readiness gaps into go-live gates. It explains security and secrets, provider live readiness, customer access and budgets, routing and fallback, support and observability, billing and commercial rules, and customer handoff. Each gate has an owner, required evidence, current evidence, approval question, status, and next step. It also shows required signoffs and rollout stages from internal live test to broader rollout.

Change Management Plan

/v1/gateway/change-management explains how to change customers, providers, or model routes without surprising a customer. It includes change types, risk level, approval owner, before-change checklist, after-change validation, rollback path, and evidence endpoints. It is not automatic rollback yet, but it is a simple operating plan for safer demos, pilots, and production discussions.

Change Request Preview

/v1/gateway/change-request-preview previews one planned customer, provider, or model route change before saving it. It returns would_apply=false, risk level, approval owner, affected provider, affected models or customers, before-change checklist, after-change validation, rollback path, and evidence endpoints. This helps a customer understand that a gateway needs change control. It is not automatic approval, automatic rollback, config version history, or a production change workflow.

Preview Output	Simple Meaning
would_apply=false	The endpoint explains the change but does not save it.
risk level	Shows whether the change is low, medium, or high risk.
approval owner	Shows who should approve before the change happens.
affected models or customers	Shows who or what may be impacted.
rollback path	Shows how the team would recover if the change causes a problem.

Data Governance Review

/v1/gateway/data-governance explains what happens to prompt data, response data, logs, customer gateway keys, and provider secrets. It covers prompt handling, provider secret handling, customer key handling, logging and retention, sensitive data preview, customer visibility, and production gaps. It is not a compliance certification. It is a simple checklist for customer trust discussions before real production traffic.

Question	Why it matters
Can prompts be logged?	Support teams need traces, but prompts may contain sensitive customer data.
How long are logs kept?	Customers need retention and deletion rules before production use.
Who can see request details?	Admin access must be limited and auditable.
Where are provider keys stored?	Provider secrets must live in a secret manager, not customer-facing JSON.
What sensitive data is blocked or redacted?	The gateway needs clear customer-specific data handling rules.

Security Review

/v1/gateway/security-review explains the main security questions a customer will ask before trusting one gateway with many AI providers. It covers customer gateway key risk, provider secret risk, sensitive prompt data in logs, wrong customer or model access, unsafe admin changes, current prototype controls, production controls still needed, and go-live security gates. It is a simple threat model for customer conversations, not a penetration test, SOC 2 report, legal compliance review, or production approval.

Risk	Current prototype control	Production control needed
Customer key leak	Allowed models, request limits, budgets, rotate and disable access	Real keyless access, approvals, emergency disable, and alerts
Provider secret exposure	Provider records use environment variable names and messages, avoid sensitive secrets	Secret management, restricted operators, and secrets rotation
Sensitive prompt data in logs	Safety preview can show obvious sensitive patterns	Retention rules, masking, deletion workflow, and customer-specific controls
Wrong model access	Customer allowed_models and access matrix	Tenant isolation tests, export controls, and stronger policy review
Unsafe admin change	Lifecycle endpoints create audit events and change approvals	Approval workflow, staged rollout, config history, and automated rollback

Operations Runbook

/v1/gateway/operations-runbook explains what to watch after the gateway is used by a customer, who should act, and what evidence to collect before changing routes or promises. It covers SLO-style targets, daily checks, alert actions, owner responsibilities, current signals, and evidence endpoints. It is a simple operations runbook for customer pilots, not a legal SLA or full monitoring system.

Area	What it means
SLO-style targets	Track health, error rate, latency, and budget before promising legal SLA terms.
Daily checks	Review health, provider status, customer errors, budgets, and security alerts.
Alert actions	Use first actions and customer-safe wording for provider, request, budget, and security signals.
Ownership	Name support, gateway, platform, and business owners before a pilot starts.
Evidence	Use alerts, provider health, request detail, customer reports, invoice preview, and support policy.

Model Evaluation Plan

/v1/gateway/evaluation-plan explains how to compare models before routing real customer traffic. It covers task quality, reliability, latency, cost, safety and data handling, fallback behavior, sample evaluation prompts, model scorecards, and evidence endpoints. It is a simple quality plan for customer conversations, not a full benchmark platform or automatic ranking system.

Dimension	Customer question
Task quality	Does the model answer the customer's real use case well?
Reliability	Does the route work repeatedly without confusing errors?
Latency	Is the response fast enough for the customer workflow?
Cost	Is the model affordable under the customer's budget?
Safety	Can the request be sent without exposing sensitive data?
Fallback	What happens if the first provider is unavailable?

Customer Discovery Checklist

`/v1/gateway/discovery-checklist` helps before promising an OpenRouter-like gateway. It turns a broad customer idea into clear scope by asking about the customer goal, model and provider scope, tenant rules, data governance, commercial reporting, and production expectations. It also lists evidence to collect, red flags, fit assessment, and a recommended first pilot.

Area	What to clarify
Customer goal	What workflow calls the gateway first and who decides pilot success.
Provider scope	Which provider and capabilities must work first.
Tenant rules	Which customers can use which models, budgets, and keys.
Data rules	What can be logged, retained, redacted, or deleted.
Production expectation	Whether this is a demo, pilot, limited production, or full production.

Proposal Summary

`/v1/gateway/proposal-summary` turns discovery into a simple customer-facing scope note. It explains the customer problem, recommended positioning, phase one scope, what is not in phase one, customer deliverables, decision points, main risks, and recommended next steps. It is not a legal quote or final contract.

Part	Purpose
Phase one scope	Show what the first pilot should include.
Not in phase one	Avoid promising marketplace, SLA, billing, or many providers too early.
Decision points	Help the customer choose custom gateway, first provider, mock/live test, and pilot/production path.
Risks	Explain provider differences, data handling, and prototype readiness clearly.
Next steps	Move from discussion to one small technical pilot.

Recommended Roadmap

Phase	Goal	Result
1. Current prototype	Qwen-only gateway with mock dashboard	Customer can understand the concept
2. Live Qwen demo	Use Model Studio API key for real chat	Validate latency and response quality
3. Gateway controls	Persist logs, usage, model access rules, budgets, and BYOK for customer management	Enterprise-ready direction
4. More providers	Enable OpenAI, Claude, Xiaomi adapters	More complete OpenRouter-like direction
5. Production	Database, secrets, streaming, fallback, billing	Enterprise-ready platform path

Reference Documents

OpenRouter API Reference: <https://openrouter.ai/docs/api/reference/overview/>

OpenRouter Authentication: <https://openrouter.ai/docs/api-keys>

OpenRouter Limits: <https://openrouter.ai/docs/api-reference/limits/>

OpenRouter Models API: <https://openrouter.ai/docs/api/api-reference/models/get-models>

OpenRouter Model Fallbacks: <https://openrouter.ai/docs/guides/routing/model-fallbacks>

OpenRouter Provider Routing: <https://openrouter.ai/docs/guides/routing/provider-selection/>

OpenRouter BYOK: <https://openrouter.ai/docs/use-cases/byok/>

Alibaba Cloud Model Studio DashScope API Reference: <https://www.alibabacloud.com/help/doc-detail/3016809.html>

Final Recommendation

Start small and explain clearly. Use the visual dashboard to show the customer-facing API, the routing layer, and the provider adapter. Use mock mode for discussion and live Qwen mode only when real model testing is needed. Add more providers only after the customer understands the value of the gateway.